

Interaction models for surgical planning in eXtended Reality. Challenges in radiologist-surgeon communication

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ABSTRACT

Extended Reality (XR) technologies have been included in medical applications, showing benefits for patient well-being. For medical image visualization, XR offers promising benefits but also design and evaluation challenges. In this preliminary work, direct observations and interviews were conducted with two expert health workers (one surgeon and one radiologist) who used an XR technology to perform immersive anatomic visualization for surgical planning for two years. Interaction requirements and expectations were collected, which provided a basis for the position that communication during surgical planning improves if the virtual environment offers interaction models for both medical specialties. Subsequently, a technical review of medical image visualization software in PC and XR platforms was carried out to analyze the interaction techniques provided to the users. Finally, interaction models were conceived to offset the lack of interaction techniques oriented to the surgeon's needs.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

The inclusion of eXtended Reality (XR) technologies in healthcare has provided relevant contributions to various medical applications, including treating psychological and mental conditions, neurorehabilitation and guidance for invasive procedures such as surgeries, interventional procedures, endoscopy, among others [2, 3, 10]. Another promising field is immersive medical image visualization for surgical planning purposes, where the introduction of XR technologies has contributed to a better understanding of complex anatomical structures [6, 9].

Medical image visualization is a key activity of surgical planning, which activity requires cooperation between different clinical and surgical or interventional specialties. However, two specialists lead the surgical planning: (1) radiology, providing the technology and methods to acquire medical images according to the pertinence of the clinical case and (2) surgery, which requires all the patient clinical information, including image data, to elucidate the intervention method supported. The more pathology becomes anatomically complex, the more sophisticated the image technique is required. Then, in severe malformation, the abstraction of the pathology can result complex for the surgeon, even using a high-quality multi slice images. Some solutions for that scenario have been addressed, including strengthening image quality, realistic volume rendering, 3D printing, improving communication protocols between medical specialties, and using immersive medical visualization. This last one has provided promising results. However, the evaluation has been mainly approached in clinical studies without deepening the user performing the different interaction techniques [4].

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Currently, different open-source and membership medical image software allow to use of XR modules or provides a full XR solution. This feature broadened the possibilities to interact with 3D data and improving the understanding anatomical diseases. Nevertheless, some challenges related to interdisciplinary collaboration between radiology and surgery specialties acquire a key role during this planning and present different training and wisdom, which may generate a communication gap and make it difficult to understand anatomical features by the surgeon.

Additionally, most of the acknowledged XR medical image software provides tools with an interaction design similar to used at standard medical image software that uses traditional inputs/outputs (keyboard, mouse and flat display). In this way, these interaction models do not take advantage of XR technology. For this reason, in this positional paper, we question immersive interaction design by addressing the healthcare endeavor and expectations of XR technology for surgical planning, particularly the radiology and surgery specialties, because, as previously mentioned, these are key roles in this medical activity. This paper also presented a work in progress that involves the technical review of medical image visualization software, the interview and observation of the endeavor of two experienced healthcare workers, one surgeon and one radiologist. Section 2 presents the related work of this initiative, section 3 the materials and methods used, section 4 shows the main finding and preliminary results, including the surgeons' and radiologists' requirements for immersive interaction and eight interaction models designed to fulfill those needs, and section 5 discusses the actual work and future research.

2 RELATED WORK

The implementation and evaluation of interaction models and metaphors have been addressed in research, involving XR technologies for applications in different sectors. In the construction area, a study compares immersive versus traditional Building Information Modeling tools for Computer Fluid Dynamics in the context of building projects [13]. The evaluation of interaction models can include peripherals such as Internet of Things (IoT) devices for applications in home automatization [1] and smartwatch metaphor-based hand gesture interface for supporting more natural 3D navigation in mobile VR Versus traditional Mobile VR interfaces, evidencing that the smartwatch method outperformed button-click and hand-gesture mobile VR interaction [11]. For educational applications, the evaluation of interaction models helps to find better alternatives for personnel in training, as occurs with the evaluation of conceptual models of multimodal interactions in virtual reality to solve the problems encountered in 3D printing learning in a university laboratory [5] and the and evaluation of two interaction models for augmented reality interactive registration that aim to be minimally invasive to the workflow and to mimic natural manipulation of 3D object [8]. Other healthcare applications comprise the evaluation for reducing cybersickness using VR rehabilitation applications [12] and the development and evaluation of an immersive analytic system that provides interaction models adapted for investigating volumetric data with minimal invasion to the radiologist's traditional tasks [7].

3 MATERIALS AND METHODS

The findings and preliminary results presented in this work were the outcomes of fieldwork comprised of four stages performed. The first stage consists of the direct observation of healthcare personnel interacting with XR technology for surgical planning and a subsequent interview to identify their needs during surgical planning. The second phase involves selecting and reviewing software for medical image visualization to identify the features of their navigation and manipulation tools. Therefore, the third phase was the comparison between healthcare personnel needs and the technical offer for immersive anatomical visualization. Finally, a creative process to elucidate adequate interaction models suitable for future implementation in XR platforms was carried out to fulfill the gap between needs and technical offers. A detailed description of each phase is presented as follows.

Software	Platform		
	PC	XR Module	XR
Horos	✓		
Osirix Lite	✓		
3D Slicer	✓	✓	
Mimics Materialise	✓	✓	
Inobitec	✓		
Elucis			✓
Medical Holodeck			✓

Table 1: The list of medical image visualization software selected to perform the navigation and manipulation interaction techniques review.

3.1 Healthcare personnel observation and interview

Two experienced health workers were in contact with an in-house XR technology for a long period (three years). Subsequently, they were interviewed by the first author of this work, who is a biomedical engineer and Ph.D. student, regarding their immersive visualization impressions: (1) a radiologist with 24 years of experience and (2) a surgeon with 40 years experience in general and cardiovascular surgery. Both of them entered into contact with an in-house solution developed for HTC Vive to perform immersive medical image visualization towards surgical planning [9]. Their experience and performance were documented through annotations, photos and video. Finally, an interview was conducted to identify their needs and expectations to improve their experience with XR technology. Due to their clinical experience and the time involved with the XR solution, the health workers were considered sufficient to participate as subjects of the current research.

3.2 Technical review and data extraction

A scientific and technical review was performed to identify seven acknowledged medical image visualization technologies. Subsequently, the solutions were evaluated by the first author, who had four years of anatomical reconstruction experience. This process allowed to identify the tools provided for the software and compare them to the health workers' needs and expectations using immersive medical visualization (Table 1).

3.3 Interaction model conception

The authors of this work performed a creative process considering the previous comparison between health workers' expectations and the medical image software tools. The resulting metaphors were designed to improve the usability for immersive medical image visualization considering the radiologists' and surgeons' needs.

4 FINDING AND PRELIMINARY RESULTS

During the use of the XR technology, health workers were observed and interviewed about their needs and expectations. The findings show that while the radiologist considers immersive visualization as a complementary tool to express the information of the clinical case, the surgeon finds a revolutionary way to introduce himself into a virtual environment that provides the patient's anatomy and facilitates the identification of anatomical features guided by the radiologist.

This phenomenon can be explained by the fact that the radiology staff had been trained at a technical level using traditional tools, hardware and peripherals such as the mouse, keyboard and monitor, allowing them to understand three-dimensional anatomy navigating slice-by-slice. However, due to the lack of familiarity, they can feel confused about free-movement interaction techniques such as drag-and-drop in an immersive environment. By contrast, the surgeon finds it easier to use this technique than to translate over a linear movement by selecting the X, Y and Z axes, which can make them feel restricted but is more intuitive for the radiologists. The above-mentioned suggests that the immersive interaction thought for radiologists should be similar to traditional navigation and manipulation tools in 2D inputs but adapted to the possibilities of immersive virtual environments. On the other hand, interaction requirements thought of by surgeons should tend towards the natural, resembling what they would seek to do in the surgery.

Furthermore, the two health workers were asked about expectations and possibilities to improve the immersive experience: The radiologist manifests the usefulness of highlighting important anatomy features, not only marking, measuring or commenting but the possibility of changing the color of irregular structures. On the other hand, the surgeon manifests the value of having the ability to operate the virtual anatomy, in other words, to have interaction techniques for manipulating the 3D data familiarly to the operating room, and the possibility to see and control the movement of the anatomical structures. The health workers' behavior and expectations using immersive visualization can be summarized as follows.

Radiologist

- Prefers a virtual environment that provides familiarity with the traditional medical image visualization software (presence of the 2D navigation and manipulation techniques and rectilinear movements over classical immersive navigation techniques).
- Manifests that dominate classical immersive navigation techniques (E.g., Drag-and-drop) can be challenging.
- Expect intuitive tools to highlight key anatomy features without performing complex manipulation or segmentation processes.

Surgeon

- Can appropriate some classical immersive interaction techniques easily (E.g., Drag-and-drop). However, requires enough awareness elements to improve orientation.
- Expects tools and elements that make the virtual environment as close as possible to the surgery procedure.
- Expects the visualization of dynamic structures in movement.

4.1 Medical image visualization software technical review

The seven acknowledged medical image visualization software were reviewed, and some provided a module or a whole solution based on

Stakeholder	Task	Horos	Osirix Lite	3D Slicer	3D Slicer VR	Mimics M	Mimics M VR	Innobitec	Elucis	M Holodeck
Radiologist	2D navigation.	✓	✓	✓		✓		✓	✓	✓
	3D MPR navigation.	✓	✓	✓		✓		✓		
	3D Volume navigation.	✓	✓	✓	✓		✓	✓	✓	✓
	Create/delete volumetric data regions.	✓	✓	✓		✓		✓	✓	✓
Surgeon	Grasp.	✓	✓	✓		✓		✓	✓	✓
	Cut.						✓			
	Dissect.								✓	
	Suture.									✓
Both specialties	Making notes.	✓	✓	✓		✓		✓		✓
	Dynamic visualization.	✓	✓	✓		✓	✓	✓	✓	✓
	Awareness elements.	✓	✓	✓		✓		✓		✓

Table 2: Relation between technical features provided by medical image visualization software and requirements identified for radiologists and surgeons.

XR. Subsequently, each software was explored to identify its navigation and manipulation tools which were studied and decomposed according to the 3D interaction techniques classification.

Although the interaction technique review is running, it had been possible to compass the general software offer with the needs of the health workers. Table 2 shows the relation between the software data extraction considering the radiologist interaction behavior and expectations. Overall, the solutions are oriented to image processing for diagnosis purposes, accomplishing the requirements for radiology practices. However, any software fulfills the surgeons' needs as the surgery-inspired model and the intuitive navigation fields are empty.

Generally, the medical image visualization software suppliers have opted to implement VR modules, as occurs with 3D Slicer and Mimics from Materialise, to provide an additional alternative to navigate through the 3D data but with a restriction in manipulating the information. Another solution, Medical Holodeck, allows for manipulating the information and even facilitate generating 3D models. Again, however, the metaphors used to perform those tasks are almost equivalent to those implemented traditionally in PC Platforms. Finally, Elucis provide a canvas menu attached to the hand controller, similar to some immersive user interface and are currently under a usability study.

Orientation awareness is an element that requires attention in immersive visualization solutions. While PC solutions provide a user interface with features such as avatars and pivots to identify image orientation, the XR modules and software show the 3D data, which, after a while, can produce disorientation due to the lack of anatomical axis references and even the user can feel completely lost if the data is scaled in a huge proportion. XR suppliers should provide awareness elements that refer to the 3D data location with medical images and user orientation. This feature is crucial, especially for surgeons who prefer free-movement interaction techniques.

Therefore, most of the current immersive solutions are centered mainly on radiology skills rather than providing features that can properly involve medical specialties such as surgery, where the professionals are not experts in traditional 2D and 3D images and sometimes in severe malformation cannot understand all the information provided by the radiologists.

4.2 Interaction model conception

The creative process involves proposing different interaction models for each health worker, aiming to offset their needs and expectations. A total of six interaction models were conceived for the current

work in progress. The model representation consists of 3D objects rendered in Unity 3D. The models are presented as follows:

4.2.1 Radiology immersive interaction models

The following models are thought for radiologists and are then presented with a degree of similarity to traditional navigation and manipulation tools in 2D inputs but adapted to the possibilities of immersive virtual environments.

Immersive 2D navigation Model oriented to navigate slice-by-slice in the three anatomical planes. However, the interaction elements include a thumbnail representing the information volume and three "slides" for each axis, which can be adjusted using controls or hand tracking. The location of these slides is reflected on three monitors respectively.

Immersive 3D MPR navigation Multiplanar reconstruction (MPR) is frequently implemented to view cavities or structures in orientations other than traditional anatomical planes. However, this proposal uses an editable plane with 6 degrees of freedom controlled by the controller/hand-tracking and then performs a confirmation action to mark said axis (Figure 1).

Immersive 3D navigation This interaction model provides an egocentric visualization controlled by an axis-shaped element that, when colliding with the controller/hand, moves the 3D data in the expected direction. The anatomical planes can be activated to contribute to location awareness.

4.2.2 Surgeon immersive interaction models

Those interaction models were conceived towards the free-movement interaction, trying to approximate what they would seek

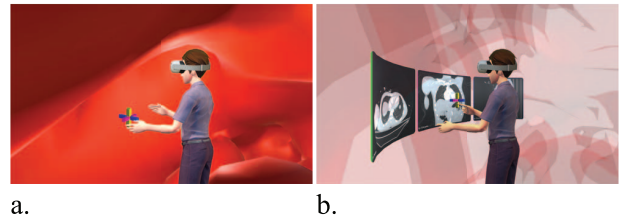


Figure 1: Representation of immersive 3D navigation: the user moves in a rectilinear manner in two views: (a) egocentric view of internal structures, (b) movement is reflected in the diagnostic images.

in the surgery environment without additional hardware as input.

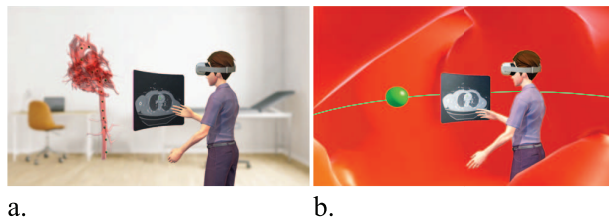


Figure 2: Spline catheterism/endoscopy: The user establishes a spline by marking nodes on the 2D images while viewing the 3D information eccentrically (a). Subsequently, the egocentric view (b) is activated in which it is possible to move the curve.

Spline Catheterism/Endoscopy This navigation mode arises from the traditional endoscopic view tool in different diagnostic imaging software using a single panel to show the 2D image, providing location awareness (Figure 2). One of the considerations observed in this modality was the challenge of navigating when the user is inside the muscle tissue. For this reason, to guarantee an exclusive path in the cardiac chambers or other cavities, the path is limited using a spline that the user creates in the two-dimensional images.

Slice and grab The Slice function, widely present in medical visualization software, allows you to “cut” a certain portion of the diagnostic image to limit the information. In a three-dimensional image, it can be useful to appreciate the interior of the anatomy. It is proposed that, for the immersive virtual environment, make “cuts” to the volume but without eliminating information, but that these become manipulable elements or shells to make it easier for the user to “explode” the part.

Sculpt ROI Creating a region of interest is essential in diagnostic image analysis, whether for area/volume measurements or visualization purposes. It is proposed to use this attribute to introduce an editable primitive in the same space as the diagnostic image to delimit its information based on this primitive.

5 FINAL REMARKS

This positional paper aims to ask about traditional interaction models present in acknowledged medical image visualization software, specialty for XR platforms, where the design possibilities are wider in terms of features for a comprehensive understanding of anatomy in an interdisciplinary team conformed by radiologists and other medical specialties. Communication challenges in the interpretation of the medical image have been addressed, through this research is expected to evaluate how immersive visualization features, specialty the capability to design virtual environments with rich interaction features, can benefit radiologist-surgeon communication. This approach has been remarkable for the health workers who enthusiastically introduced the technology in surgical planning activities.

Currently, more interaction models are presented as a partial result of this approach by analyzing the gap between acknowledged software solutions for immersive anatomical visualization and the behaviors and expectations of health workers involved in this initiative. The next step is to implement those models and conduct experimental studies to analyze the performance and perception of a mayor sample of surgeons and radiologists, aiming to generate virtual environments suitable for surgical planning where interdisciplinary healthcare professionals can participate actively.

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